Qi Huang

Female | Dec 1996
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11 Datun Road, Chaoyang District, Beijing, 100101, China

EDUCATION

National Institute of Natural Hazards, Ministry of Emergency Management of China

- Assistant Professor — Beijing, China
  Relevant research work: Hydrological extremes — July 2025-Now

EDUCATION

Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences

- PhD candidate of Physical Geography (supervisor: **Yongqiang Zhang**) — Beijing, China
  Relevant research work: Anthropogenic impact on hydrological regimes — Sep 2021-2025
- Co-cultivate post-graduate student — Beijing, China
  Relevant research work: Hydrological modelling in ungauged basins — Nov 2019-July 2021

University of Zurich

- Visiting PhD student of Geography (supervisor: **Jan Seibert**) — Zurich, Switzerland
  Relevant research work: long-term large-scale base flow change analysis — Oct 2023-Now

College of Water Resource &Hydropower, Sichuan University — Chengdu, China

- Master degree of Hydrology and Water Resources |GPA: 3.8/4.0| Sep 2018-July 2021
- Bachelor degree of Hydrology and Water Resources |GPA: 3.4/4.0| Sep 2014-June 2018

  Relevant Coursework: Hydrological Modelling; Principles of Hydrology; Hydrological Statistics; Hydrological Forecast; Hydrological Analysis; Modern Information Technique of Hydrology etc.

RESEARCH INTERESTS

Hydrological extremes
Hydrologic modelling and optimization
Runoff predictions in ungauged catchments
Large scale runoff predictions using remote sensing data

RESEARCH EXPERIENCE

July 2026-Now **Hydrological extremes**

Oct 2022-July 2026 **Global runoff signature change and attribution based on observations**

- Collecting and merging global streamflow dataset from different sources of datasets for 20,000+ catchments.
- Analyzing runoff signature change regarding different backgrounds.

Sep 2020-Dec 2022 **Estimating Vegetation Greening Influences on Runoff Signatures Using a Log-Based Weighted Ensemble Method**

- Collecting remote sensing data via Google earth engine.
- Processing massive data with R, MATLAB and ArcGIS.
- Coupling conceptualized hydrological models with a diagnostic ET model.
- Proposing an improved method for model ensemble to better simulate

runoff signatures in arid basins.

- Quantifying greening influences on runoff signatures from the aspects of not only runoff magnitude, but also high/low flow frequency/duration and flow dynamic.
- Discussing sensitivities of greening on runoff signatures.

Nov 2019-Aug 2020 **Using Remote Sensing Data-Based Hydrological Model Calibrations for Predicting Runoff in Ungauged or Poorly Gauged Catchments**

- Data pre-processing.
- Proposing nine calibration schemes for investigation.
- Analyzing modelling result at different temporal and spatial scales.
- Exploring the potential of solely using RS-AET based data for calibrating hydrological models.
- Discussing relationships between spatial characters and modelling results.

June 2019-Dec 2019 Analysis of spatial distribution characteristics of precipitation in Qilian Mountains Based on MSWEP.

Mar 2020-Aug 2019 Spatial and Temporal Analyzation of Several Precipitation Products.

June 2017-June 2018 Runoff simulation in the upper reach of Yalong river based on XAJ model.

Jan 2017-Dec 2017 Study on response of climate change and human activities on runoff.

PUBLICATIONS

Water Resources Research 2026

Huang, Q., & Zhang, Y. (2026). Disentangling the Impacts of Changes in Climate and Vegetation on Hydrological Processes Across 2,252 Global Catchments. Water Resources Research, 62(7), e2025WR043326. https://doi.org/10.1029/2025WR043326

- **Journal of Hydrology: X** 2025

Huang, Q., Jan, S., Zhang, Y.(2025) Coherent changes in global high and low flows, Journal of Hydrology:X, 30, 100212.

- **Catena** 2024
- Huang, Q., Zhang, Y., Wei H. (2024). Can direct CMIP6 model simulations reproduce mean annual historical streamflow change? Catena, 235. 10.1016/j.catena.2023.107650
- **Journal of Hydrology: Regional Studies** 2024

Huang, Q., Zhang, Y., Li, C., and Ma, N.(2024) Quantifying propagation effects of climate and vegetation changes on evapotranspiration and streamflow signatures in yarlung-tsangpo river basin, Journal of Hydrology: Regional Studies, 56, 102015.

- **Water Resources Research** 2022

Huang, Q., Zhang, Y., Ma, N., &Post, D. (2022). Estimating vegetation greening influences on runoff signatures using a log-based weighted ensemble method. Water Resources Research, 58, e2022WR032492. 10.1029/2022WR032492

- **Water Resources Research** 2020

Huang, Q., Qin, G., Zhang, Y., Tang, Q., Liu, C., Xia, J., et al. (2020). Using remote sensing data-based hydrological model calibrations for predicting runoff in ungauged or poorly gauged catchments. Water Resources Research, 56, e2020WR028205. 10.1029/2020WR028205

Developed models

HBV-PML

Teaching Experience

Teaching Assistant, Sichuan University Courses: Engineering Hydrology Mar 2019- June 2019

- Assisting students in course studying and preparing weekly examinations for students;
- Collecting feedbacks from students and professors to improve the teaching quality.

Honor & Awards

Excellent Graduation Thesis of IGSNRR	Oct 2021
Excellent Graduate student of IGSNRR	Oct 2021
Excellent Graduation Thesis of Sichuan University	Oct 2021
Excellent Graduate student of Sichuan University	Oct 2021
CYWater 2020 Young Scientist Best Paper Award: Outstanding Paper	Dec 2020
Excellent Postgraduate student of Sichuan University	Oct 2019
Excellent Graduation Thesis of Sichuan University	Oct 2018
National Encouragement scholarship	Oct 2017
First prize of "The 7th Mathor Cup National Mathematical Modeling"	July 2017
First prize of "The 14th May Day Mathematical Modeling League"	June 2017

Relevant Skills

Programming: R, MATLAB, Google Earth Engine, Visual Basic

Models: Conceptualized hydrological models;
Penman-Monteith-Leuning Version 2 Model;
HBV-PML

Software: ArcGIS, Microsoft Visio, Origin, SPSS, Office, HEC-RAS

Language: College English Test-6: 530

Website: https://www.researchgate.net/profile/Qi-Huang-75 ; https://kaye-hq.github.io/homepage/experience/